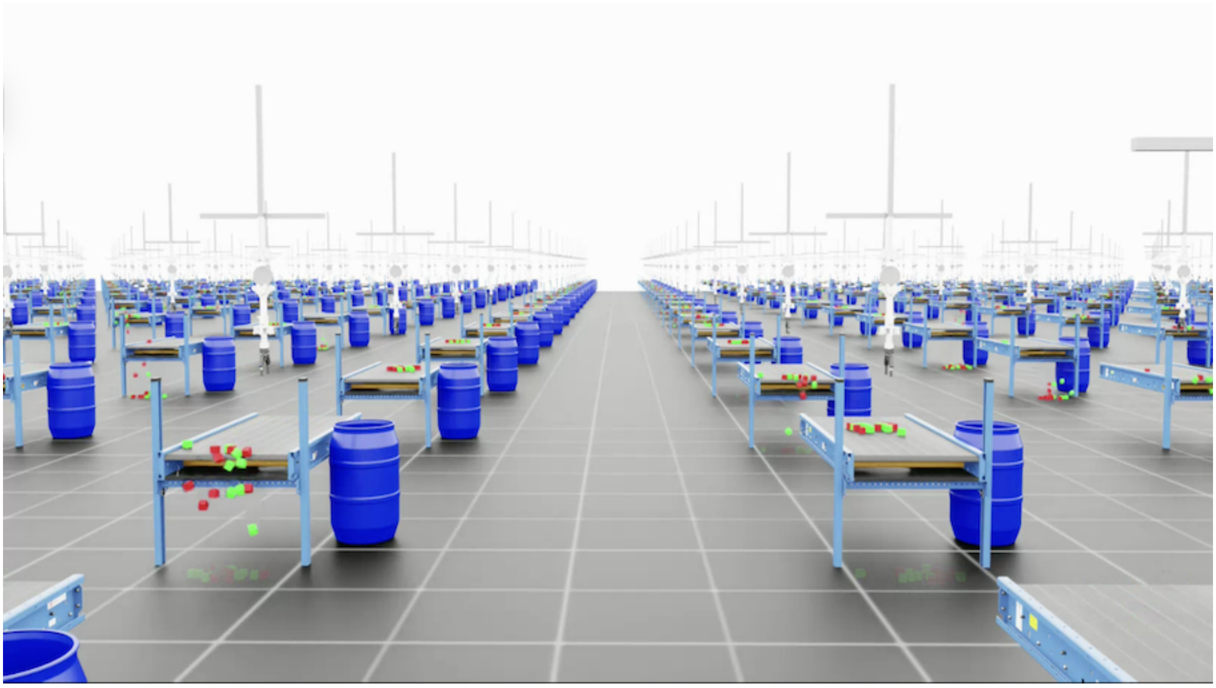
Multi-Agent Reinforcement Learning for Condition-Based Textile Sorting with a Tendon-Driven and Collaborative Robot System

Master Thesis im Studiengang Computational Engineering M. Sc.

Friedrich-Alexander-Universität Erlangen-Nürnberg
Lehrstuhl für Fertigungsautomatisierung und Produktionssystematik
Prof. Dr.-Ing. J. Franke



Bearbeiter: Georg Meyer Matrikelnr.: 22791103

Betreuer: Prof. Dr.-Ing. J. Franke
Dipl.-Ing. M. Landgraf

Abgabetermin: TBD

Bearbeitungszeit: x Monate

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Erlangen, den 12. März 2026

Georg Meyer

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List of Abbreviations

API	Application Programming Interface
CAD	Computer-Aided Design
CPU	Central Processing Unit
DoF	Degree of Freedom
DR	Domain Randomization
dVRK	da Vinci Research Kit
FEM	Finite Element Method
FK	Forward Kinematics
GAE	Generalised Advantage Estimation
GPU	Graphics Processing Unit
IK	Inverse Kinematics
IL	Imitation Learning
MDP	Markov Decision Process
MJCF	MuJoCo XML Format
MPC	Model Predictive Control
MPM	Material Point Method
PBD	Position-Based Dynamics
PD	Proportional-Derivative
PGS	Projected Gauss–Seidel
PID	Proportional-Integral-Derivative
POMDP	Partially Observable Markov Decision Process
PPO	Proximal Policy Optimization
RGB-D	Red-Green-Blue plus Depth
RL	Reinforcement Learning
ROS	Robot Operating System
SDF	Signed Distance Field
TGS	Temporal Gauss–Seidel
URDF	Unified Robot Description Format
USD	Universal Scene Description
VBD	Vertex Block Descent
XML	Extensible Markup Language
XPBD	Extended Position-Based Dynamics

List of Symbols

$\mathcal{A}$	Action space
$\hat{A}_t$	Advantage estimator
ϵ	Clipping parameter (PPO)
γ	Discount factor
h	Simulation time step
$J(\theta)$	Expected discounted return
k_j	Stiffness parameter (PBD)
$\mathbf{M}$	Mass matrix
π_θ	Policy (parameterized by θ)
$r(s_t, a_t)$	Reward function
$\mathcal{S}$	State space
θ	Policy parameters

1 Introduction

1.1 Motivation and Context

The global fashion and textile industry generates substantial volumes of post-consumer waste, with a large fraction ending up in landfill or incineration rather than being recovered for reuse or recycling. Within the European Union, regulatory pressure to increase textile recycling rates is growing, with separate collection of textile waste becoming mandatory in member states. A key bottleneck in the recycling pipeline is the sorting step: incoming textile waste must be classified by material composition, garment type, and condition (e.g., intact, minor damage, major damage) before it can be routed to appropriate downstream processes such as fiber-to-fiber recycling, downcycling, or second-hand resale. Currently, this sorting is performed predominantly by manual labor, which is slow, ergonomically demanding, and difficult to scale.

Automating textile sorting with robotic systems is a promising approach to increasing throughput and consistency, but it presents significant technical challenges. Garments are deformable objects that exhibit highly variable shapes, complex self-interactions, and contact dynamics that differ fundamentally from the rigid-object manipulation scenarios for which most industrial robots are designed [?]. Reliable textile sorting requires not only picking individual items from a mixed waste stream but also manipulating them into configurations that allow visual inspection — for example, stretching a garment flat to expose damage — and then routing each item to the correct output bin based on the inspection result. This sequence of pick, inspect, and sort naturally involves multiple manipulation primitives and can benefit from coordination between multiple robot arms operating on the same workpiece.

Reinforcement Learning offers a data-driven approach to learning such complex manipulation behaviors. Rather than manually designing controllers for each manipulation primitive, Reinforcement Learning enables policies to be discovered through trial-and-error interaction with the task environment, guided by a reward signal that encodes the desired outcome [?, ?]. When extended to *multi-agent* settings, Reinforcement Learning can learn coordinated policies for teams of robots that must share a workspace, hand off workpieces, and jointly optimize a system-level objective.

Modern Graphics Processing Unit-accelerated physics simulators have become essential enablers for this approach. By running thousands of environment replicas in parallel, they provide the sample throughput required for stable policy learning while avoiding the cost and risk of training on physical hardware [?]. NVIDIA Isaac Sim, with its PhysX 5 physics backend and the associated IsaacLab framework, provides an integrated toolchain that combines high-fidelity rigid-body and deformable-object simulation with a modular Markov Decision Process-based task interface suitable for large-scale Reinforcement Learning training [?, ?]. Recent developments in the Newton physics engine further extend this capability to realistic cloth simulation using Vertex Block Descent (Vertex Block Descent) [?, ?], making simulation-based development of cloth manipulation policies increasingly practical.

This thesis is the second part of a two-part project conducted at the Institute for Factory Automation and Production Systems (FAPS) at Friedrich-Alexander-Universität Erlangen-Nürnberg. The preceding project thesis (Meyer, 2026) established a validated simulation of the 5-Degree of Freedom tendon-driven robot and demonstrated Reinforcement Learning-based control for reach and rigid-object place tasks in Isaac Sim. The present work extends that foundation to the full textile sorting pipeline: integrating deformable cloth simulation, adding a second collaborative robot arm, implementing camera-based damage classification, training coordinated multi-agent policies, and preparing the system for sim-to-real transfer.

1.2 Problem Statement and Objectives

Condition-based textile sorting from mixed waste streams poses a multi-faceted robotic manipulation problem that goes substantially beyond standard pick-and-place. The task requires a system that can (i) selectively retrieve target garments (e.g., T-shirts) from a moving conveyor while ignoring non-target objects, (ii) present each garment in a configuration suitable for visual inspection (e.g., stretched flat), (iii) classify the garment's condition through camera-based analysis, and (iv) route the item to the appropriate output bin. Each of these sub-tasks involves contact-rich interaction with deformable objects whose behavior is difficult to model analytically [?], and the full pipeline requires coordination between multiple robotic agents operating in a shared workspace.

The robotic system considered in this thesis consists of two manipulators. The first is the tendon-driven 5-Degree of Freedom pick robot validated in the project thesis: a 2-Degree of Freedom prismatic positioning stage, a 3-Degree of Freedom tendon-driven arm, and a Robotiq 2F-140 gripper, based on the design by Klein [?]. The second is a collaborative sorting arm that receives the garment from the pick robot, stretches it for camera-based inspection, and deposits it into a condition-specific bin. The two robots must coordinate their actions — timing the handoff, avoiding collisions, and jointly optimizing the sorting throughput and accuracy — which naturally frames the problem as a multi-agent Reinforcement Learning task.

A critical technical challenge is the simulation of cloth dynamics with sufficient fidelity for policy learning. Deformable cloth introduces a high-dimensional state space (many particles/vertices), complex self-collisions and folding behavior, and sensitivity to material parameters such as bending stiffness, friction, and damping [**?**, **?**]. Policies trained on inaccurate cloth models risk exploiting simulation artifacts that do not transfer to real textiles, making the *reality gap* a central concern [**?**]. Domain randomization over cloth and environment parameters is therefore essential for developing robust policies [**?**, **?**].

The thesis pursues the following objectives:

1. **Conveyor-based sorting task — single agent (PG-1):** Extend the validated pick-and-place pipeline from the project thesis to a full sorting task: selective retrieval of target items (T-shirts) from a mixed waste stream on a moving conveyor while ignoring non-target objects. Train Proximal Policy Optimization policies with selectivity encoded in the reward function and evaluate sorting accuracy, false-pick rate, and missed-pick rate.

2. **Cloth simulation integration (PG-2):** Integrate deformable cloth simulation into the Isaac Sim environment using PhysX particle cloth (Position-Based Dynamics) or Newton thin deformables (Vertex Block Descent) [**?**, **?**]. Model T-shirt objects with realistic material properties and validate cloth behavior through qualitative and quantitative benchmarks. Apply domain randomization over cloth parameters to support robust policy learning.

3. **Second robot arm integration (PG-3):** Add a second collaborative robot arm to the simulation scene for cloth stretching and presentation. Define the arm's kinematic configuration, placement, and workspace to complement the tendon-driven picker. Implement the stretching primitive: the second arm grasps a garment edge and extends it to a flat, camera-visible configuration.

4. **Camera-based damage classification (PG-4):** Integrate a simulated camera system for garment inspection. Develop or integrate a classification model that assesses textile damage degree (e.g., intact, minor damage, major damage, unusable) from the stretched cloth image. Train or fine-tune the classifier on synthetic data with domain randomization.

5. **Multi-agent coordination (PG-5):** Formulate the full pipeline as a multi-agent Markov Decision Process: the tendon-driven robot picks and hands off, the sorting arm stretches, inspects, and sorts. Design the multi-agent observation and action spaces, inter-agent communication or shared reward structure, and coordination protocol. Train coordinated policies and evaluate end-to-end sorting performance.

6. **Sim-to-real transfer preparation (PG-6):** Assess transfer robustness through systematic domain randomization (conveyor speed, cloth properties, sensor noise, lighting) [**?**, **?**]. Define the Robot Operating System 2 integration architecture for physical deployment: joint state publishers, policy inference nodes, camera pipelines, and safety constraints. Identify and document remaining gaps for real-world operation.

7. **Evaluation and benchmarking (PG-7):** Define and apply comprehensive evaluation metrics: sorting accuracy by damage class, cycle time, false/missed classification rates, cloth coverage during inspection, and multi-agent coordination efficiency. Compare against single-agent baselines and ablation studies (e.g., without stretching, without classification).

1.3 Thesis Outline

This thesis is structured as follows. Chapter 2 provides the background required for this work, beginning with a summary of the foundation established in the project thesis (tendon-driven robot model, simulation environment, and Reinforcement Learning pipeline) and then covering the additional topics central to the master thesis: cloth and deformable object simulation, multi-agent Reinforcement Learning, vision-based classification, prior work on robotic cloth manipulation and sorting, and sim-to-real transfer strategies.

Chapter 3 identifies the specific research gap: the absence of an integrated, simulation-based multi-agent system that combines tendon-driven manipulation, deformable cloth handling, camera-based condition classification, and coordinated sorting within a single Reinforcement Learning-trainable environment.

Chapter 4 describes the methodological approach, including the cloth simulation setup, the second robot arm integration, the damage classification pipeline, the multi-agent Markov Decision Process formulation, the Reinforcement Learning training procedures, and the domain randomization strategy for sim-to-real robustness.

Chapter 5 presents the experimental results across all objectives: single-agent sorting performance, cloth simulation fidelity, multi-agent coordination metrics, classification accuracy, and the end-to-end sorting evaluation.

Chapter 6 discusses the implications and limitations of the approach, including cloth simulation fidelity and its effect on policy transfer, the effectiveness of multi-agent coordination versus sequential single-agent execution, and the remaining challenges for real-world deployment.

Chapter 7 concludes the thesis, summarizes the contributions, and identifies directions for future work, including physical deployment via the Robot Operating System 2 integration pathway.

2 Background

2.1 Summary of Foundation (Project Thesis)

2.2 Cloth and Deformable Object Simulation

Deformable object simulation relaxes the rigid-body assumption by allowing the object's shape to change under load. This introduces two coupled complexities: the state dimension increases dramatically (many vertices/particles instead of a single rigid transform), and contacts become configuration-dependent because deformation continuously alters the set of potential contact points and normals [?, ?]. For textiles, this coupling is particularly pronounced: applied forces not only translate the object but also create folds, wrinkles, and contact-rich self-interactions that alter future dynamics. These properties are repeatedly identified as key contributors to the reality gap in robotic cloth manipulation [?].

Common modelling approaches. Three modelling families are most relevant in robotics simulation. First, mass–spring(-damper) models discretize cloth as a mesh whose vertices are connected by springs encoding stretch and (optionally) bending penalties; such models can be made stable with implicit integration techniques but require careful parameterization and can be numerically stiff [?]. Second, particle-based constraint methods represent cloth by particles and enforce desired behavior via geometric (holonomic) constraints, which can be solved iteratively with high robustness under collisions; Position-Based Dynamics and its variants are canonical examples [?, ?]. Third, continuum formulations discretize elastic energy and constitutive laws (e.g., in Finite Element Method) or use hybrid particle–grid methods (e.g., Material Point Method), providing a pathway to parameters with clearer physical interpretation at increased computational cost [?, ?].

Challenges specific to cloth dynamics. Cloth exhibits high effective Degree of Freedom, frequent self-collision, and strong sensitivity to material parameters such as bending stiffness, friction, and damping. Robust self-contact and friction handling is widely recognized as a hard requirement for stable cloth simulation, particularly under large deformations and repeated collisions [?, ?]. From a learning perspective, this sensitivity implies that small modelling errors (e.g., in friction or damping) can change grasp success rates, sliding behavior, and fold formation, which in turn can qualitatively alter the learned policy's strategy and its transferability to hardware [?, ?].

2.2.1 PBD — PhysX Particle Cloth

PhysX provides particle-based simulation through `PxPBDParticleSystem`, whose solver is explicitly described as using Position-Based Dynamics for particle–particle dynamics and as supporting phenomena including cloth, inflatables, and other deformable behaviors [?, ?]. In Omniverse, particle simulation is Graphics Processing Unit-accelerated and is documented as supporting fluids, granular media, and cloth, including interaction with rigid bodies and articulations [?]. At the feature level, particle cloth is treated as a particle-based Position-Based Dynamics mass–spring-style system and is marked as deprecated in favor of newer deformable-body features, which highlights that "cloth" in Isaac Sim can refer to multiple, evolving solver families rather than a single canonical model [?, ?].

Time-step structure and iterative projection. In standard Position-Based Dynamics, the deformable is represented by particle positions $\mathbf{x}_i \in \mathbb{R}^3$ with masses m_i; desired behavior is enforced by a set of constraints $C_j(\mathbf{x})$ (equalities or inequalities), typically solved by iterative projection in a Gauss–Seidel fashion [?]. A schematic time step consists of: (i) updating velocities under external forces, (ii) predicting positions by an explicit step, (iii) iteratively projecting predicted positions to reduce constraint violations, and (iv) updating velocities from the corrected positions (and applying damping if configured) [?]. For a constraint $C_j(\mathbf{x}) = 0$, linearization yields a correction direction based on the constraint gradient. Using the inverse mass matrix $\mathbf{M}^{-1}$ (diagonal in the particle basis), one common update can be written as [?, ?]:

$$\Delta\mathbf{x} = k_j \cdot s_j \cdot \mathbf{M}^{-1}\nabla C_j(\mathbf{x}), \qquad s_j = \frac{-C_j(\mathbf{x})}{\nabla C_j(\mathbf{x})^\top \mathbf{M}^{-1}\nabla C_j(\mathbf{x})}, \tag{2.1}$$

where $k_j \in [0,1]$ is a user-defined stiffness-like scaling applied per constraint.

Effective stiffness and dependence on h and iteration count. A central practical implication of the iterative projection view is that the *effective* stiffness of constraints depends on numerical settings. Increasing the number of projection iterations typically yields stiffer constraint satisfaction at fixed h, while increasing h tends to reduce stability and can require either more iterations or smaller substeps to maintain comparable behavior [?, ?, ?]. This is beneficial for robustness and controllability, but it complicates mapping simulator parameters to physically measured material properties (e.g., bending stiffness in SI units), which is a recurring issue when calibrating cloth simulation for sim-to-real transfer [?, ?].

XPBD compliance formulation. Extended Position-Based Dynamics extends Position-Based Dynamics by introducing a compliance parameter and an explicit Lagrange multiplier state to reduce time-step dependence and recover meaningful constraint force estimates [?]. For compliance $\alpha \geq 0$ and time step h, the incremental multiplier update for a constraint can be expressed as [?]:

$$\Delta\lambda = \frac{-C(\mathbf{x}) - \frac{\alpha}{h^2}\lambda}{\sum_i w_i \|\nabla_{\mathbf{x}_i} C(\mathbf{x})\|^2 + \frac{\alpha}{h^2}}, \qquad \Delta\mathbf{x}_i = w_i \nabla_{\mathbf{x}_i} C(\mathbf{x}) \Delta\lambda, \tag{2.2}$$

with inverse masses $w_i = 1/m_i$. This formulation decouples the user parameter α from the iteration count in a way that improves interpretability of stiffness-like behavior across different h and solver settings [?].

Why Position-Based Dynamics is used in PhysX for particle cloth. Position-Based Dynamics is well-aligned with Graphics Processing Unit execution because constraints can be partitioned into independent sets and processed in parallel; graph coloring is a common strategy to schedule constraint batches without write conflicts on shared particles [?]. Omniverse documentation explicitly emphasizes Graphics Processing Unit-accelerated Position-Based Dynamics particles and their interaction with rigid bodies and articulations, which is essential for cloth manipulation scenes in robotics [?]. In addition, the PhysX particle system API is designed around a dedicated Position-Based Dynamics solver type (`PxPBDParticleSystem`), reinforcing that particle cloth is treated as a specific, robust, high-throughput model family within the PhysX multiphysics portfolio [?, ?].

2.2.2 VBD — Newton Thin Deformables

While PhysX particle cloth is based on explicit prediction and iterative position projection, implicit methods aim to improve stability under stiffness and large time steps by solving a time-discrete optimization problem. Vertex Block Descent is introduced as a block coordinate descent solver for the variational form of implicit Euler time integration, targeting convergence to the implicit Euler solution while maintaining unconditional stability under limited iteration budgets [?].

Variational implicit Euler and global energy minimization. Let $\mathbf{x}_t$ denote positions, $\mathbf{v}_t$ velocities, h the time step, and $\mathbf{M}$ the mass matrix. In the variational formulation, the next positions are obtained by minimizing a global energy that combines inertia and potential energy [?]:

$$\mathbf{x}_{t+1} = \arg\min_{\mathbf{x}} G(\mathbf{x}), \qquad G(\mathbf{x}) = \frac{1}{2h^2}\|\mathbf{x} - \mathbf{y}\|_{\mathbf{M}}^2 + E(\mathbf{x}), \tag{2.3}$$

where $\mathbf{y} = \mathbf{x}_t + h\mathbf{v}_t + h^2\mathbf{a}_{\text{ext}}$ and $E(\mathbf{x})$ is the total potential energy. Velocities follow from the implicit Euler relationship $\mathbf{v}_{t+1} = (\mathbf{x}_{t+1} - \mathbf{x}_t)/h$ [?].

Vertex-level block coordinate descent and local Newton step. Vertex Block Descent solves the global problem by iterating over vertices and updating one vertex at a time while temporarily holding the others fixed, yielding a vertex-level Gauss–Seidel pattern with favorable parallelization via vertex coloring [?]. For vertex i, a local energy is minimized:

$$\mathbf{x}_i \leftarrow \arg\min_{\mathbf{x}_i} G_i(\mathbf{x}), \qquad G_i(\mathbf{x}) = \frac{m_i}{2h^2}\|\mathbf{x}_i - \mathbf{y}_i\|^2 + \sum_{j \in F_i} E_j(\mathbf{x}), \tag{2.4}$$

where F_i is the set of force/energy elements incident to vertex i. Because G_i has only three degrees of freedom, a local Newton step is formed by solving a 3×3 linear system per vertex [?]:

$$\mathbf{H}_i \Delta\mathbf{x}_i = \mathbf{f}_i, \tag{2.5}$$

with $\mathbf{H}_i$ the local Hessian and $\mathbf{f}_i$ the local gradient/force term.

Advantages for stiff, contact-rich thin deformables. The primary advantages highlighted for Vertex Block Descent are: (i) unconditional stability inherited from the energy descent property, (ii) convergence toward the implicit Euler solution as iterations increase, and (iii) Graphics Processing Unit-oriented parallelization via vertex coloring with small, localized linear solves [?]. These properties are attractive for cloth-like thin deformables, which often exhibit near-inextensible stretch (high stiffness) combined with softer bending and frequent contact events; implicit/variational formulations can tolerate such stiffness ratios at larger time steps than explicit solvers with comparable iteration budgets [?].

Newton in the robotics simulation ecosystem. Newton is presented as an open-source, Graphics Processing Unit-accelerated physics engine designed for robotics simulation and robot learning, built on NVIDIA Warp and intended to integrate with robot learning frameworks such as IsaacLab [?, ?]. In NVIDIA's robotics communication, Newton is explicitly associated with thin-deformable simulation via a Vertex Block Descent solver and with multiphysics capabilities that include implicit Material Point Method and other solver components, motivated by stability and performance for learning workflows [?, ?]. Accordingly, Vertex Block Descent is positioned as a solver choice when increased stability under stiff cloth behavior and larger time steps is required, complementing the high-throughput Position-Based Dynamics-based particle cloth pathway in PhysX-centric stacks [?, ?].

2.2.3 Comparison: PBD vs. VBD for Cloth in Reinforcement Learning Workflows

Mathematical targets and implications. The core distinction is the optimization target. Position-Based Dynamics advances dynamics by predicting positions and then *projecting* them to reduce constraint violations, where the resulting behavior depends on solver scheduling and iteration count, and where constraint "stiffness" is typically not time-step invariant without extensions [?, ?]. Vertex Block Descent, in contrast, directly minimizes the variational implicit Euler energy and therefore targets a principled implicit integration objective; stability is maintained by construction via energy descent, and increased iteration budgets refine the approximation toward the implicit solution [?].

Practical tradeoffs for learning-centric simulation. For Reinforcement Learning, Position-Based Dynamics-based particle cloth is attractive because it is simple, robust under frequent contacts, and aligned with Graphics Processing Unit-parallel execution through constraint batching and coloring strategies; this supports the high environment throughput required for policy optimization [?, ?, ?]. However, because effective stiffness depends on h and iteration count, parameter tuning can be difficult to interpret physically and can amplify sim-to-real sensitivity if policies rely on artefacts (e.g., overly damped folds or under-resolved frictional sticking) [?, ?]. Vertex Block Descent is attractive when stability under stiffness, large time steps, and mixed constraint types is prioritized, particularly for systems that combine near-inextensible stretch with softer bending; such regimes are common in cloth manipulation and can be challenging for explicit projection methods at fixed compute budgets [?, ?].

Relation to the Omniverse ecosystem and thesis choice. In Omniverse/Isaac Sim, particle cloth is documented as deprecated in favor of newer deformable-body features, indicating an ongoing transition toward more general deformable schemas and alternative solver families [?, ?]. For the experiments in this thesis, the PhysX/Position-Based Dynamics particle cloth pathway is motivated by Graphics Processing Unit throughput and robustness for large-scale training in Isaac Sim 5.1, whereas Newton/Vertex Block Descent is conceptually aligned with scenarios where greater stability under stiff cloth behavior and larger time steps is required (e.g., when cloth parameters or contact conditions become numerically challenging for projection-based solvers) [?, ?, ?].

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